

Road Traffic Accident Severity Analysis & Prediction

Project Stakeholder: Eko Hospital

Project Objective

Analyze 100 traffic accident records to identify accident patterns, hotspots, time-of-day trends, weather conditions, vehicle types, traffic density, and accident severity, then develop a classification model that can provide structured severity-prediction insights relevant to Eko Hospital.

1. What Was Done

The project covered the following analytical activities:

- Data preprocessing and exploratory analysis of **100 accident records**.
- Accident-location/hotspot analysis to identify areas with higher accident frequency.
- Time-of-day analysis across **4 periods**: Morning, Afternoon, Evening, and Night.
- Weather-condition, vehicle-type, traffic-density, and accident-severity comparisons.
- Random Forest classification using **5 input variables**: Time of Day, Weather Conditions, Vehicle Type, Traffic Density, and Accident Hour.
- Model evaluation using accuracy, classification metrics, and a confusion matrix.

2. Quantified Project Scope

Measure	Value
Accident records analyzed	100
Severity categories	3 — Minor, Moderate, Severe
Predictive input variables	5
Time-of-day groups	4 — Morning, Afternoon, Evening, Night
Traffic-density groups	3 — Low, Medium, High
Classification model	Random Forest

3. Who the Analysis Was Done For

The project is framed for **Eko Hospital**. The accident findings are translated into potential operational insights for emergency-response planning and resource allocation.

4. Results

The analysis produced a structured view of accident frequency and severity across location, time, weather, vehicle type, and traffic-density dimensions. The Random Forest workflow provides a repeatable classification approach for accident-severity prediction using the five available accident characteristics.

5. Potential Impact for Eko Hospital

The resulting insights can support Eko Hospital in planning for potential emergency demand, reviewing accident-related patterns, and considering how available emergency resources may be aligned with observed accident conditions and hotspots.

6. Important Data Interpretation

The available project materials do **not** provide a verified numerical model-accuracy result or a measured percentage improvement in hospital response. Accordingly, this updated project does not claim an unsupported accuracy percentage, improvement percentage, or clinical outcome.

7. CV-Aligned Project Statement

Road Traffic Accident Severity Analysis & Prediction — Eko Hospital — analyzed **100 traffic accident records** to identify accident patterns, hotspots, time-of-day trends, and severity-related factors; developed a **Random Forest classification model** using **5 accident characteristics** — time of day, weather conditions, vehicle type, traffic density, and accident hour —

and evaluated its predictions using accuracy, classification metrics, and a confusion matrix; translated the findings into data-driven insights to support Eko Hospital's **emergency-response planning and resource-allocation decisions**.

Project Deliverable

A Jupyter Notebook containing the preserved analytical workflow, visualizations, classification model, evaluation outputs, and the updated Eko Hospital project framing.